

TinyML Visual Alarm

On-device person detection with real-time notification

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Smart alarm systems SoA

Risks and limitations of Cloud surveillance



- Continuous raw video stream (high Mbps)
- Variable latency, susceptible of uplink drops



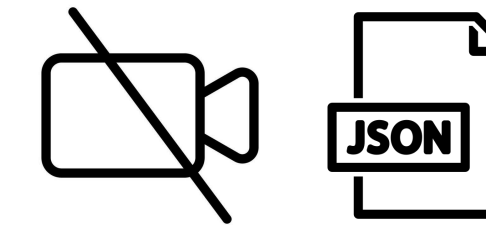
- Severe privacy risk.
- Third party storage vulnerabilities.

Smart edge alarm approach

Pushing inference to the data source



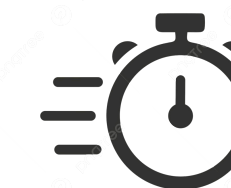
- Zero bandwidth for video, kbps for JSON.



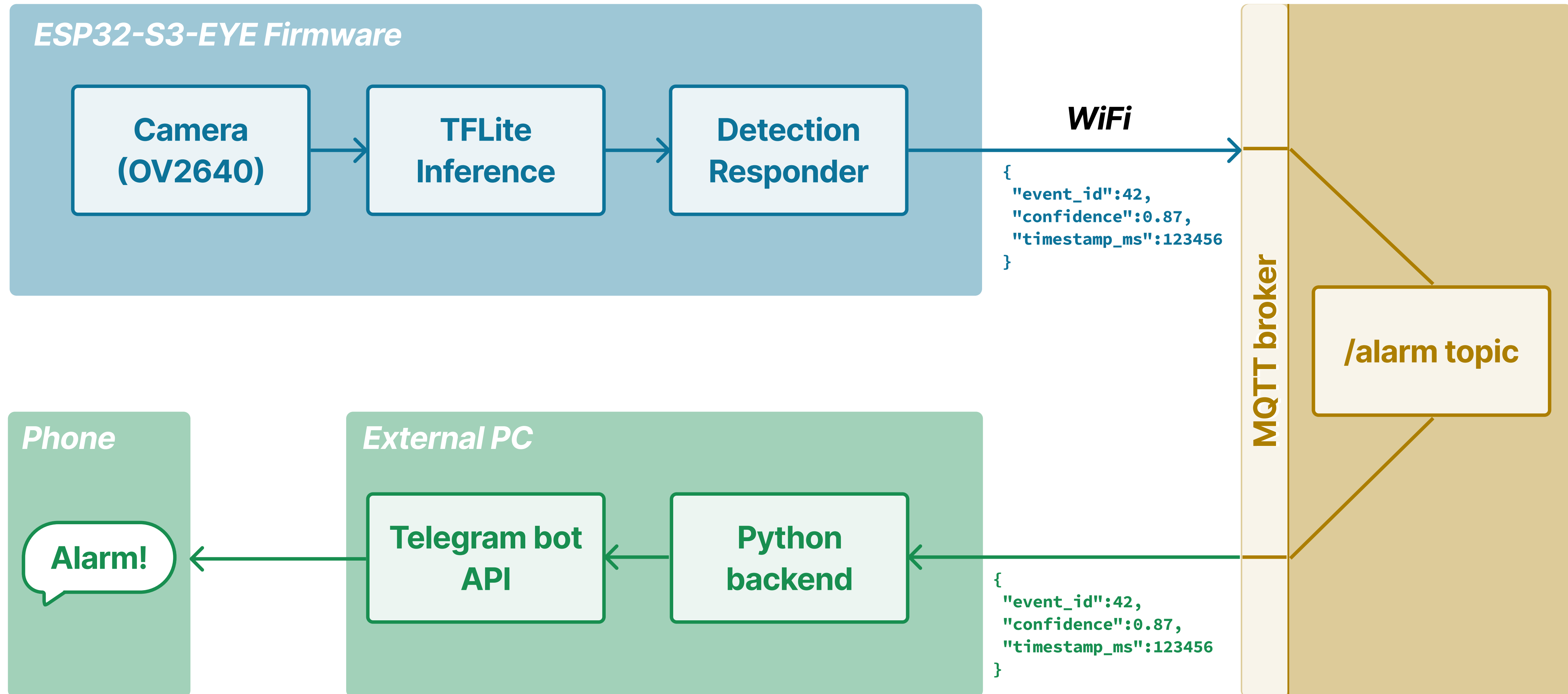
- Absolute privacy. Raw pixels never leave the device.



- Deterministic latency (not connection-related).



Architecture

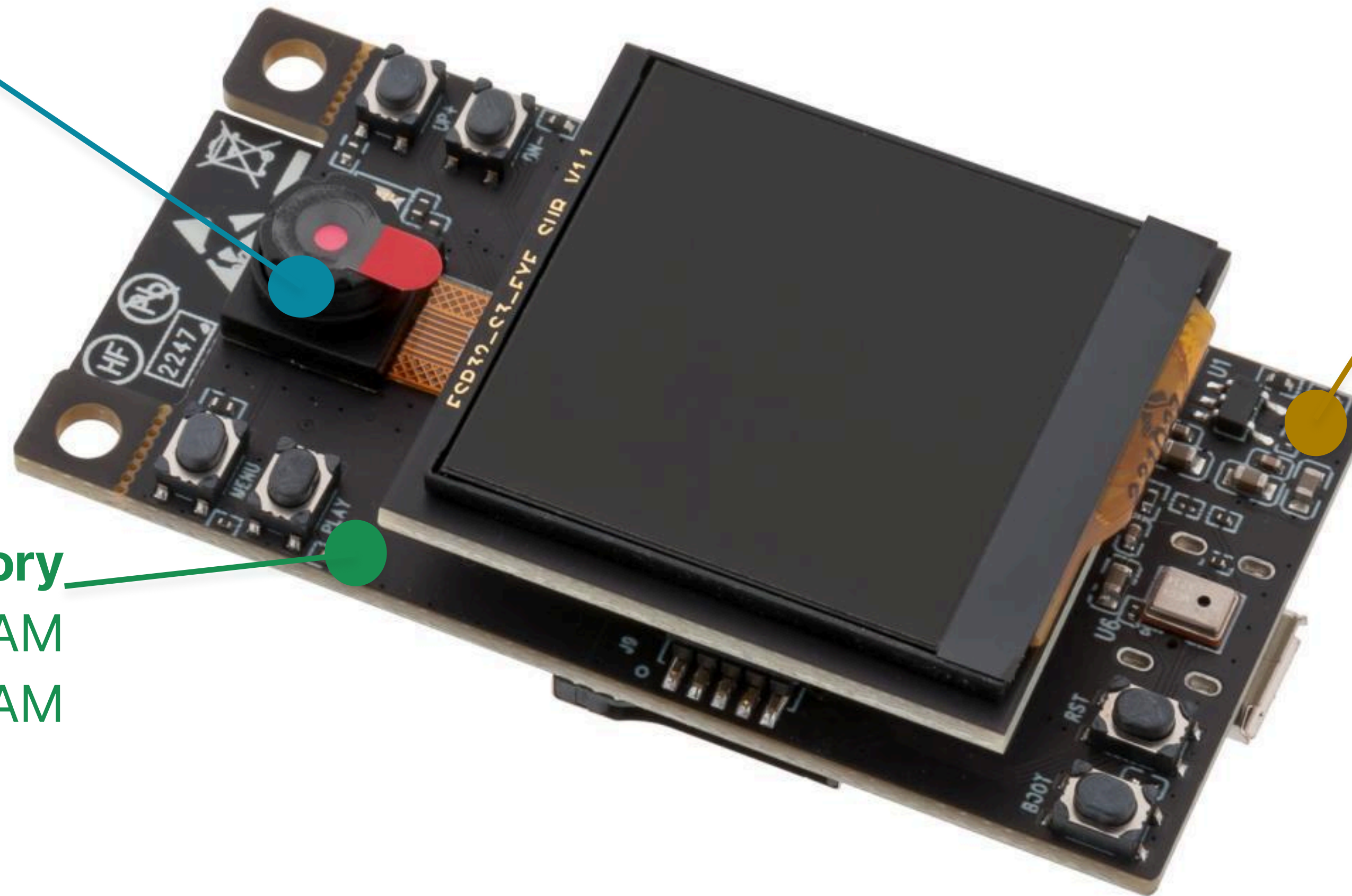


Hardware: ESP32-S3-EYE

OV2630 Camera
1600×1200 (2MP)
resolution

Dual-core Xtensa LX7 @ 240 MHz

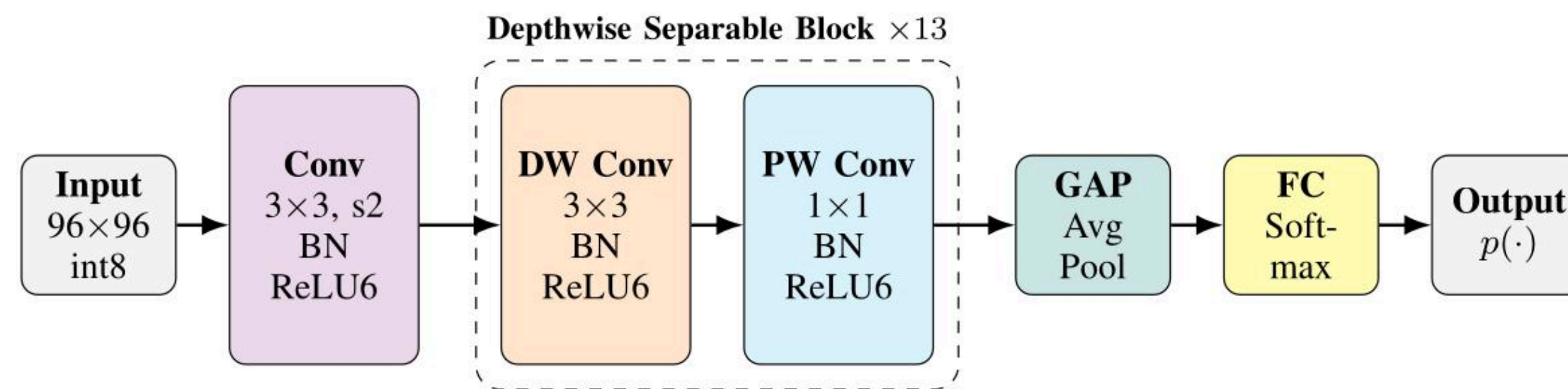
Memory
512 kB on-chip SRAM
8 MB PSRAM



MobileNetV1* (2017)

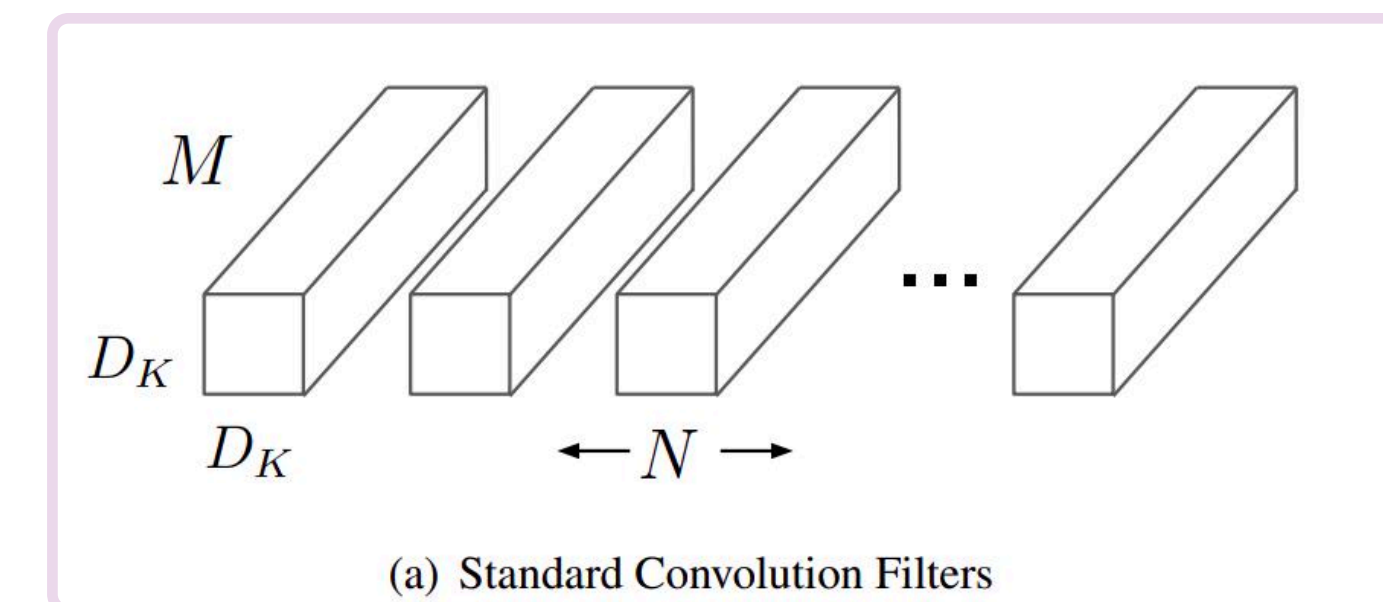
*A. Howard et al., "MobileNets: Efficient CNNs for Mobile Vision Applications," arXiv:1704.04861, 2017.

Main feature: Depthwise separable blocks



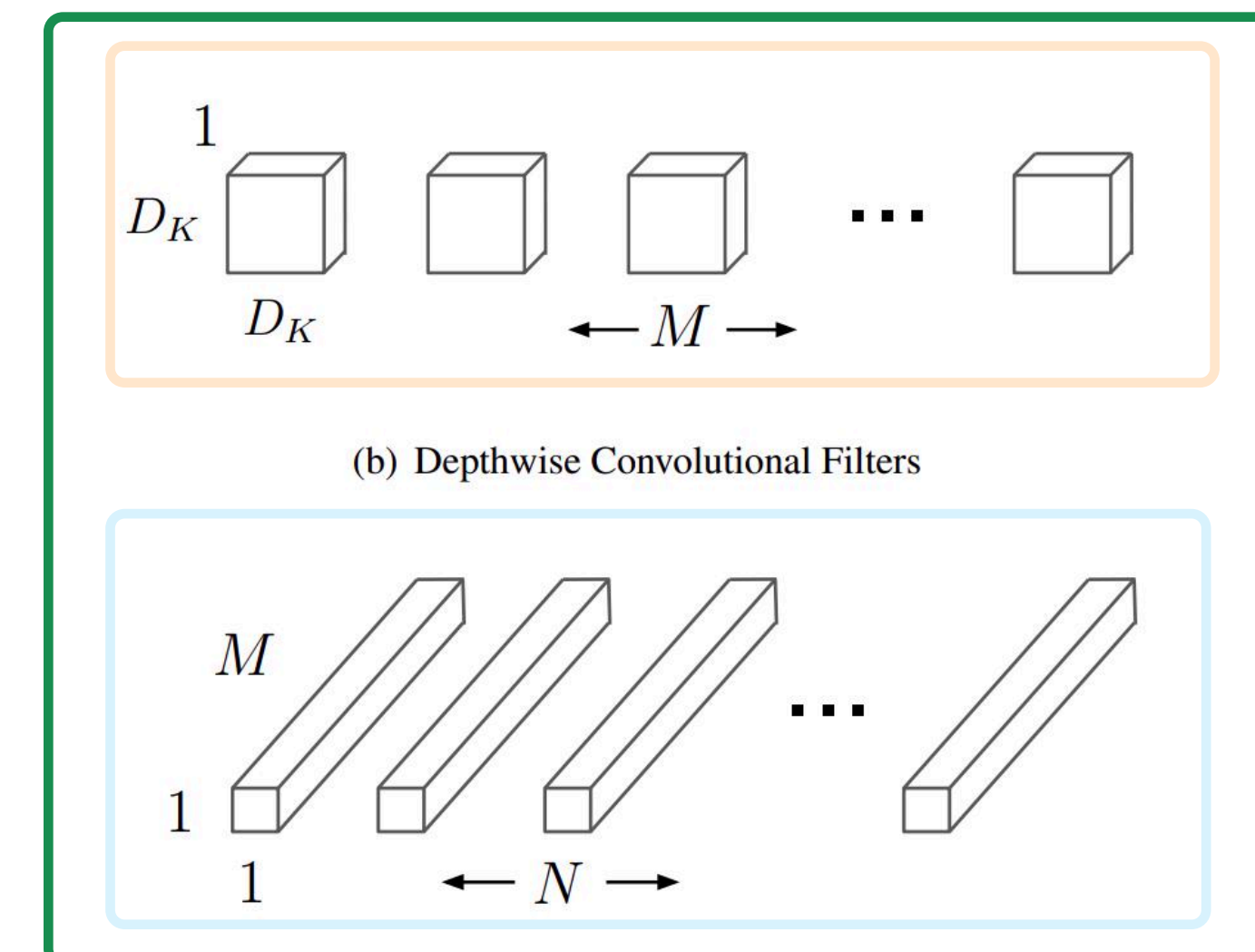
Design

- Depth multiplier $\alpha=0.25$ scales channels uniformly
- Spatial downsampling via strided depthwise convolution (no max pooling)
- # Params: 213,272

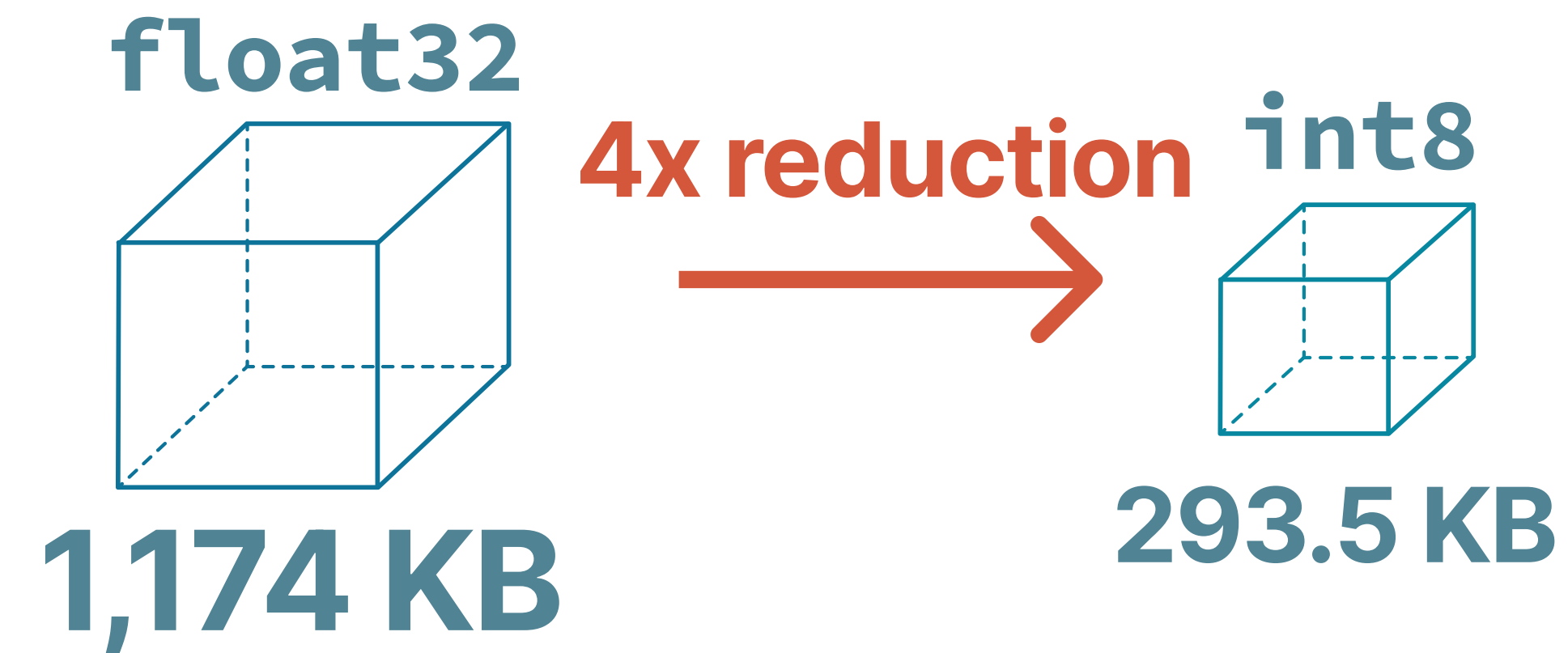


VS

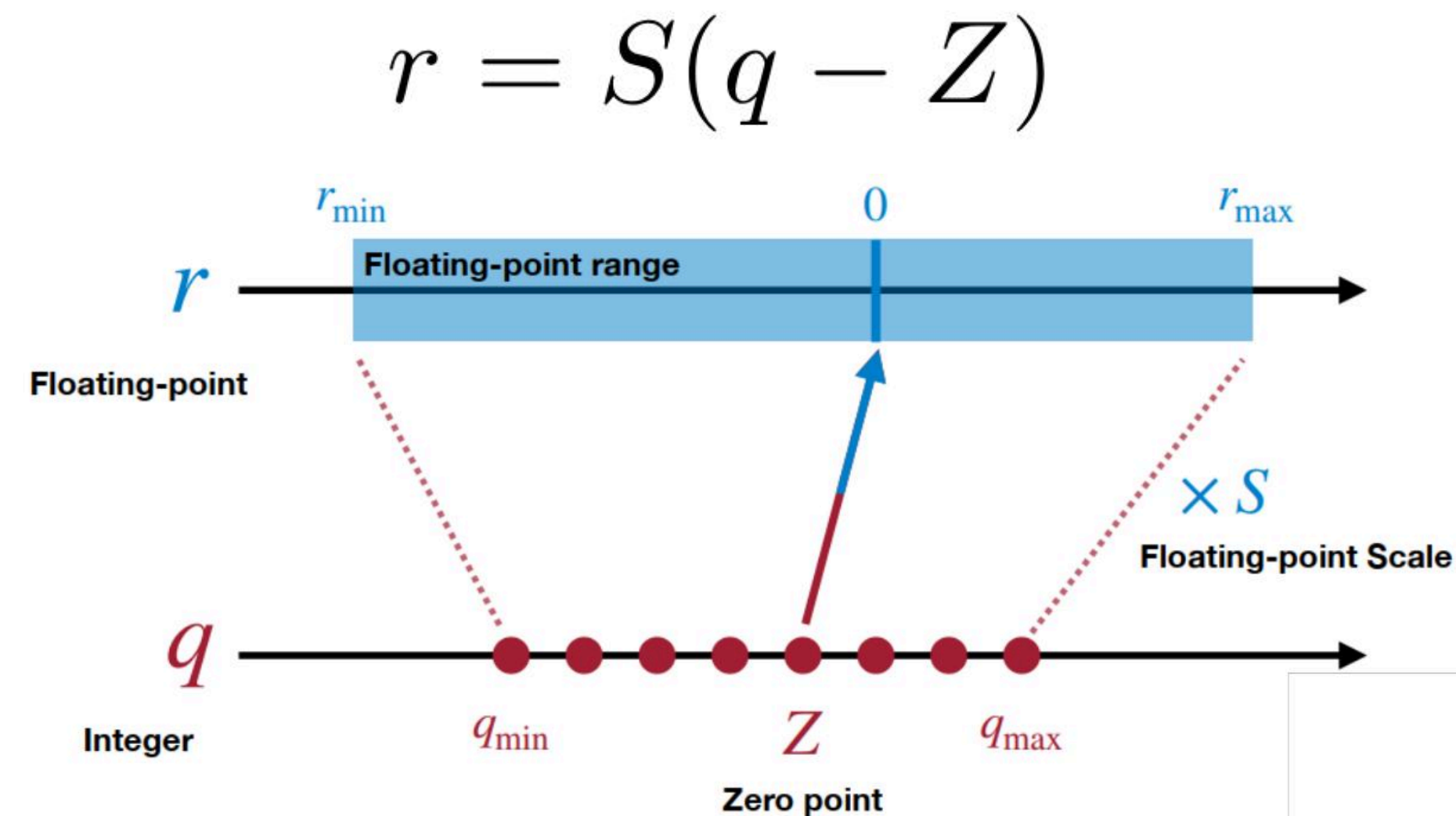
9x MACs reduction



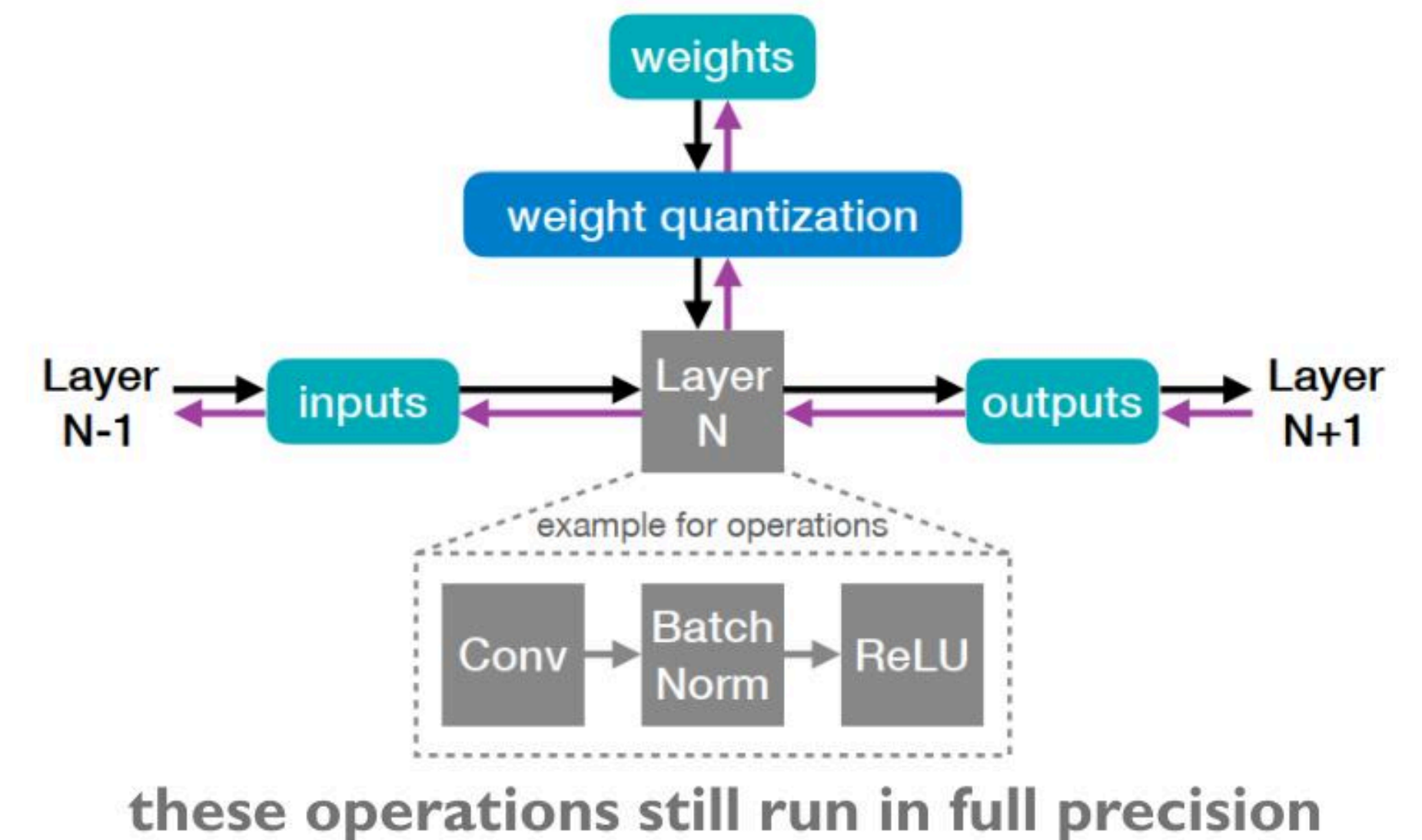
Model compression



Stock model: Post-training linear quantization (PTQ)



Custom model: Quantization-aware training (QAT)



Model training (QAT)

Phase 1 - 5 epochs, $lr = 1e-3$

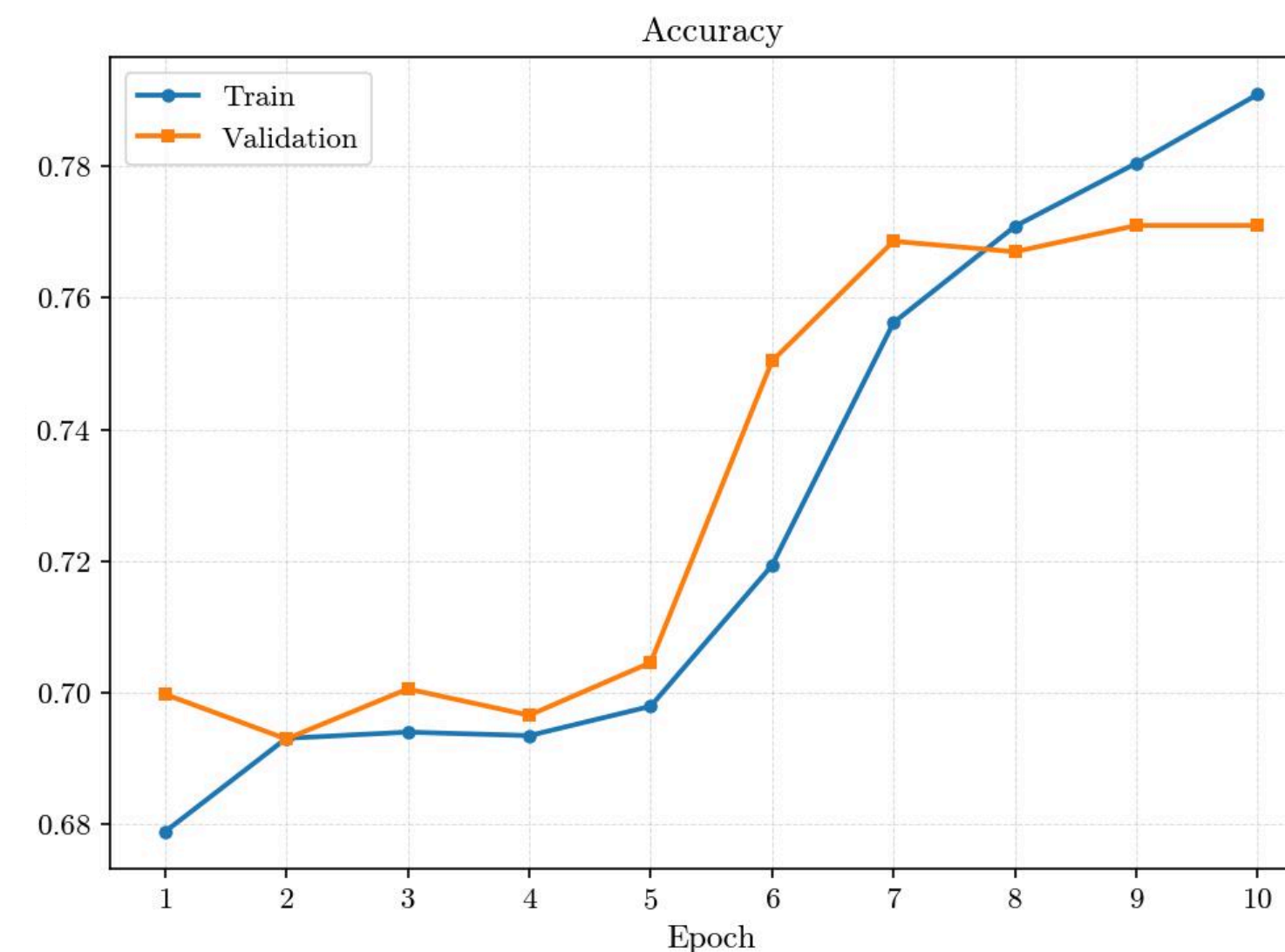
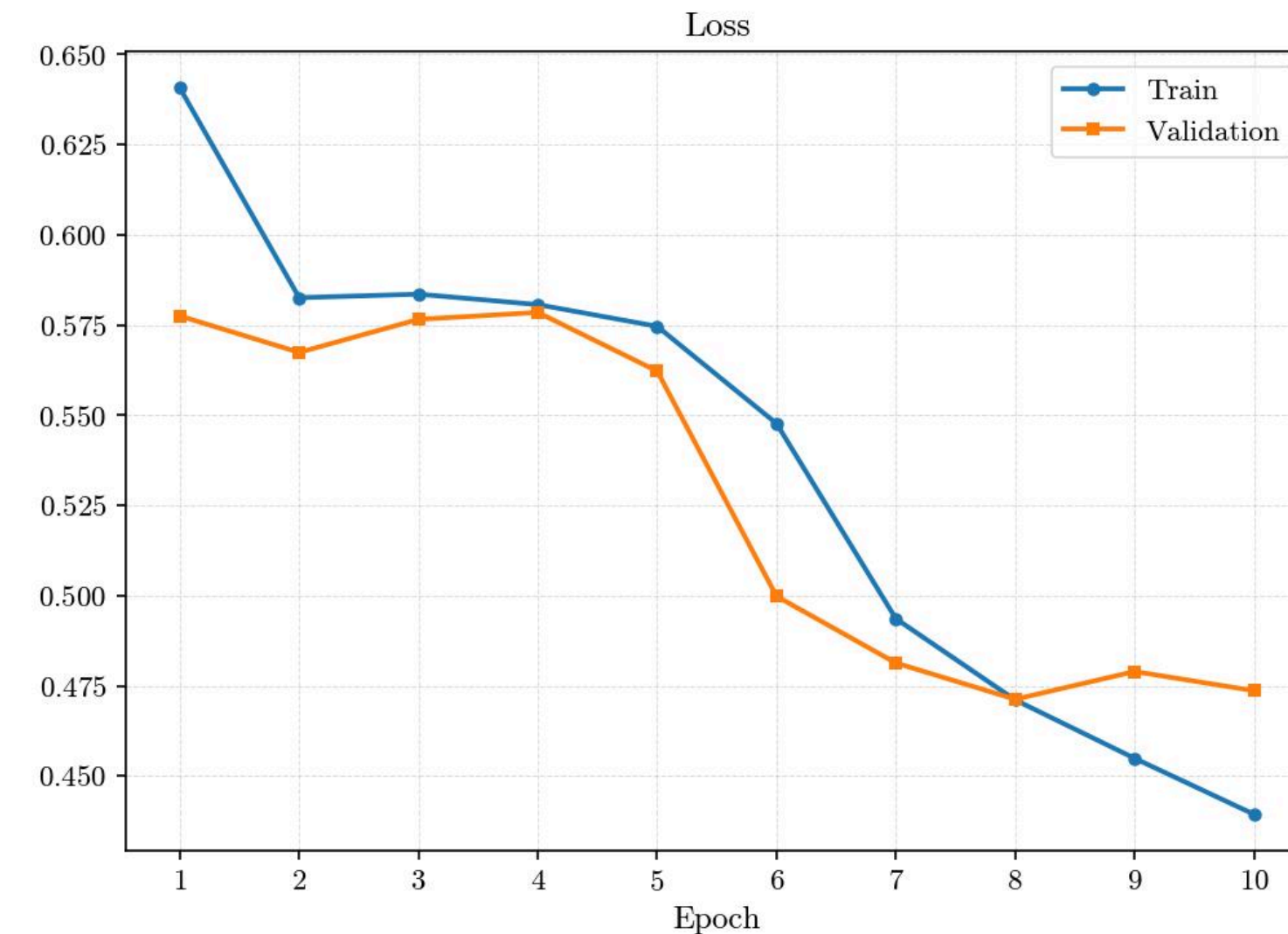
Backbone with ImageNet weights
frozen, train head only

Phase 2 - 5 epochs, $lr = 1e-4$

Full fine-tuning (end-to-end)

QAT - 3 epochs, $lr = 1e-5$

Fake-quant operators inserted
Weights adapt to 8-bit rounding

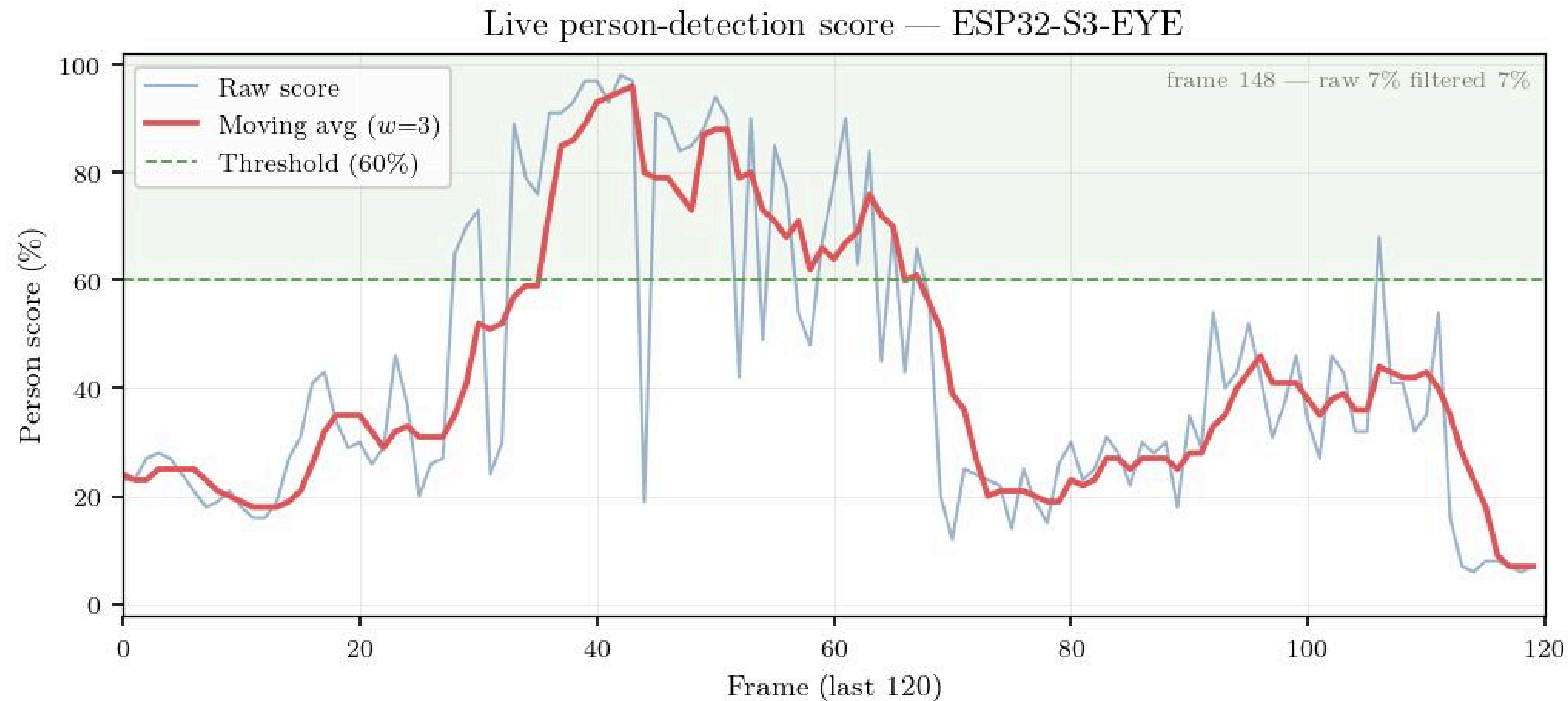


Model	AUC	Δ AUC	Acc.
Float32	0.859	—	77.3%
PTQ int8	0.855	-0.004	76.1%
QAT int8 (ours)	0.862	+0.003	77.4%

Callbacks:

EarlyStopping (patience=4)
ReduceLROnPlateau ($\times 0.5$, patience=2)
ModelCheckpoint (best val acc)

Detection logic



Design choices

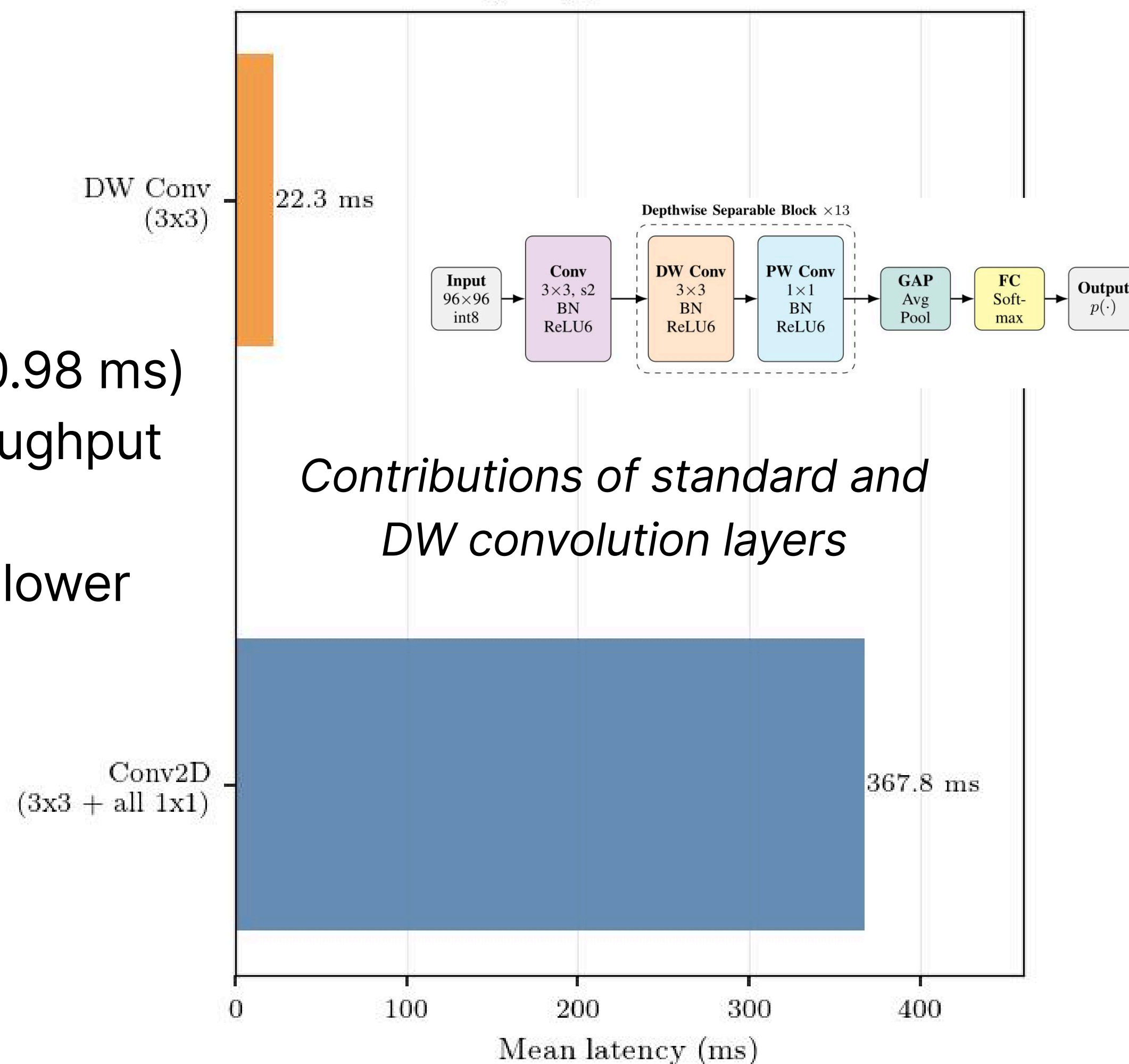
- 60% trigger threshold
- 10s cooldown (prevents alarm flooding)
- 3-frames window ($\sim 1s$)

Note: these are the default values, but this parameters can be tuned from *menuconfig*

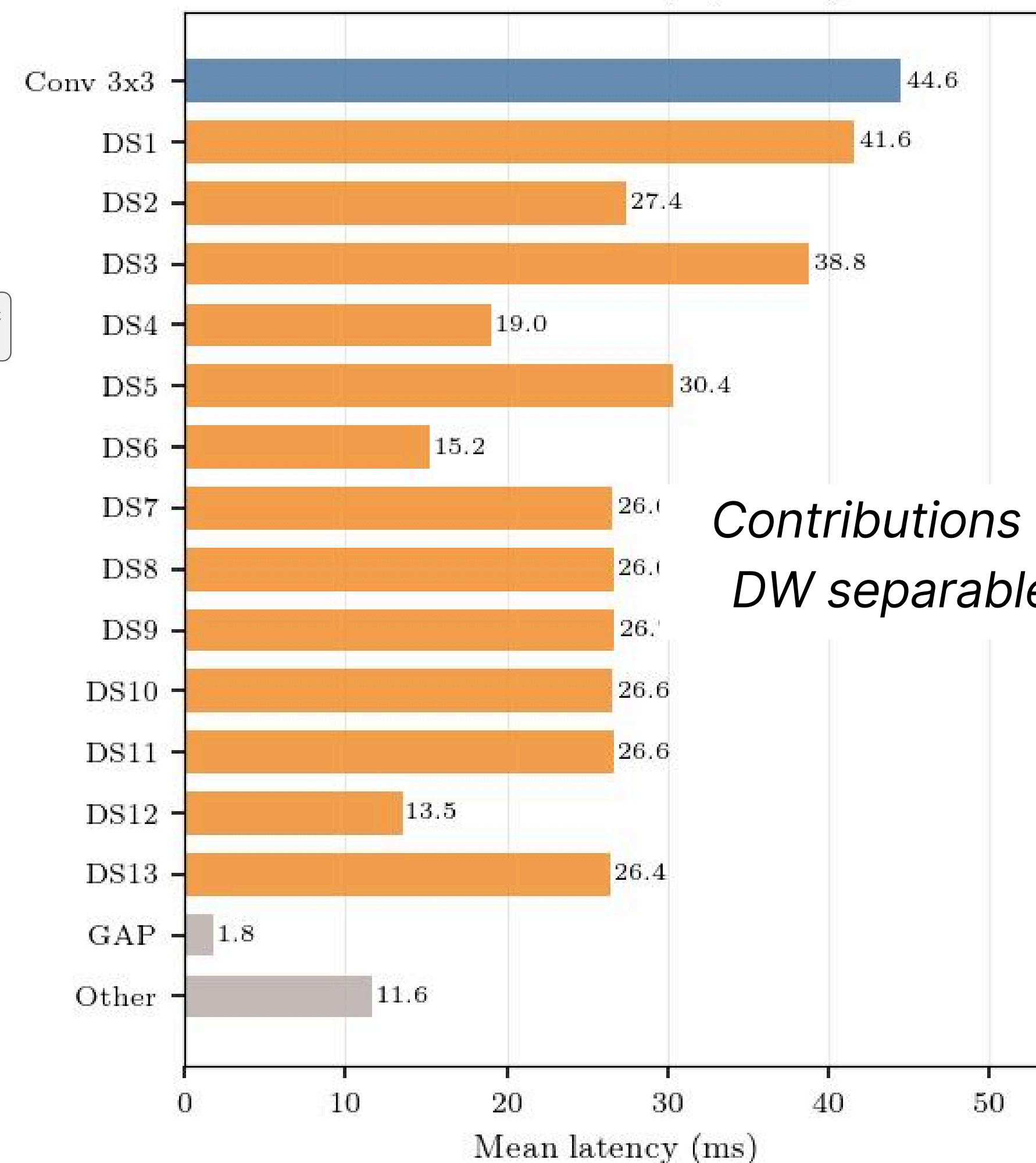
Metrics I: Hardware latency

Live Inference Latency — ESP32-S3-EYE

Layer-type breakdown



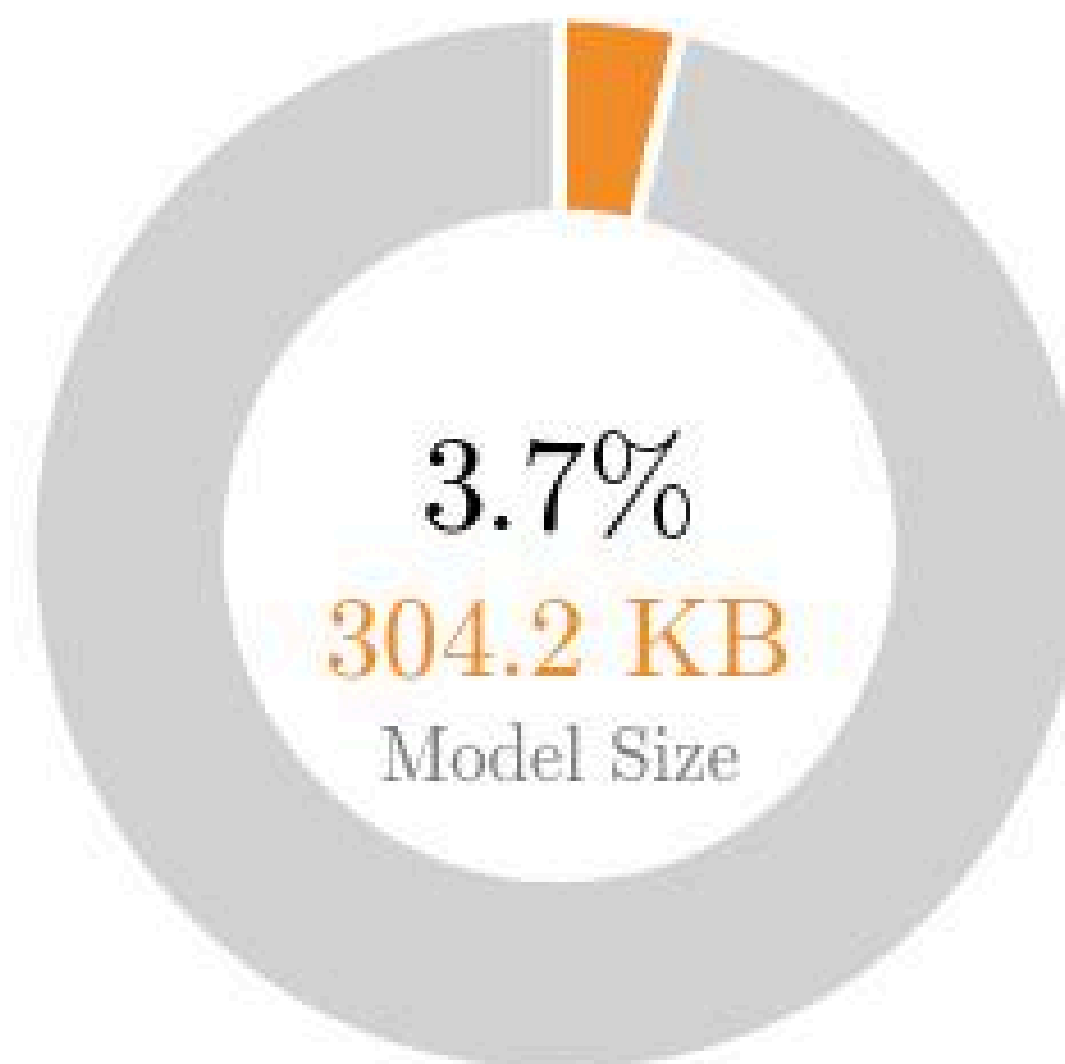
Per-block latency (n=101)



- 403.5 ms (± 0.98 ms)
- Effective throughput 2.48 FPS
- Early blocks slower (larger maps)

Metrics II: Memory & compute footprint

Storage / Flash



8 MB Total Flash

Runtime / PSRAM



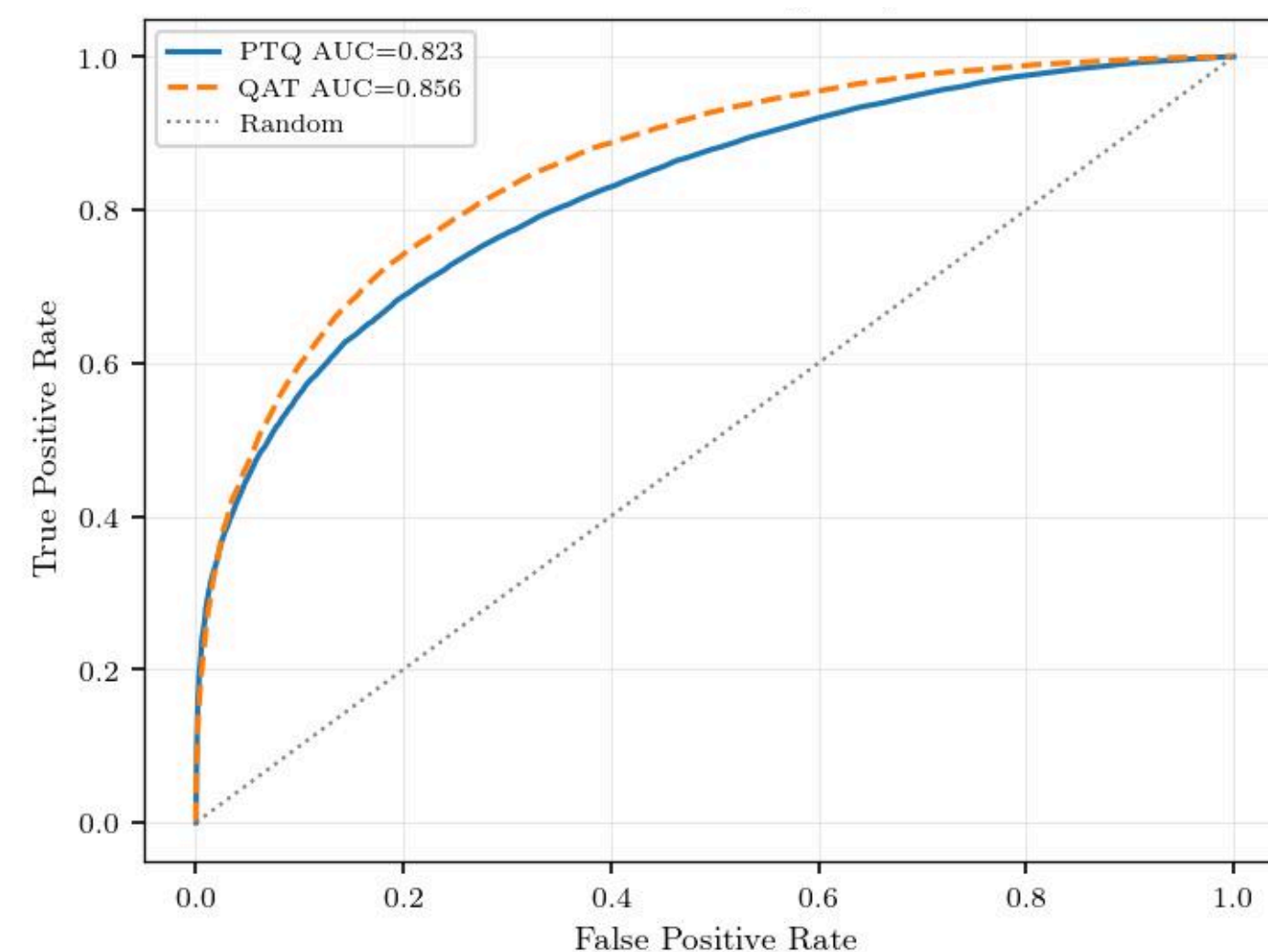
8 MB PSRAM

Throughput:
7.5 MMACs /
15 MFlops

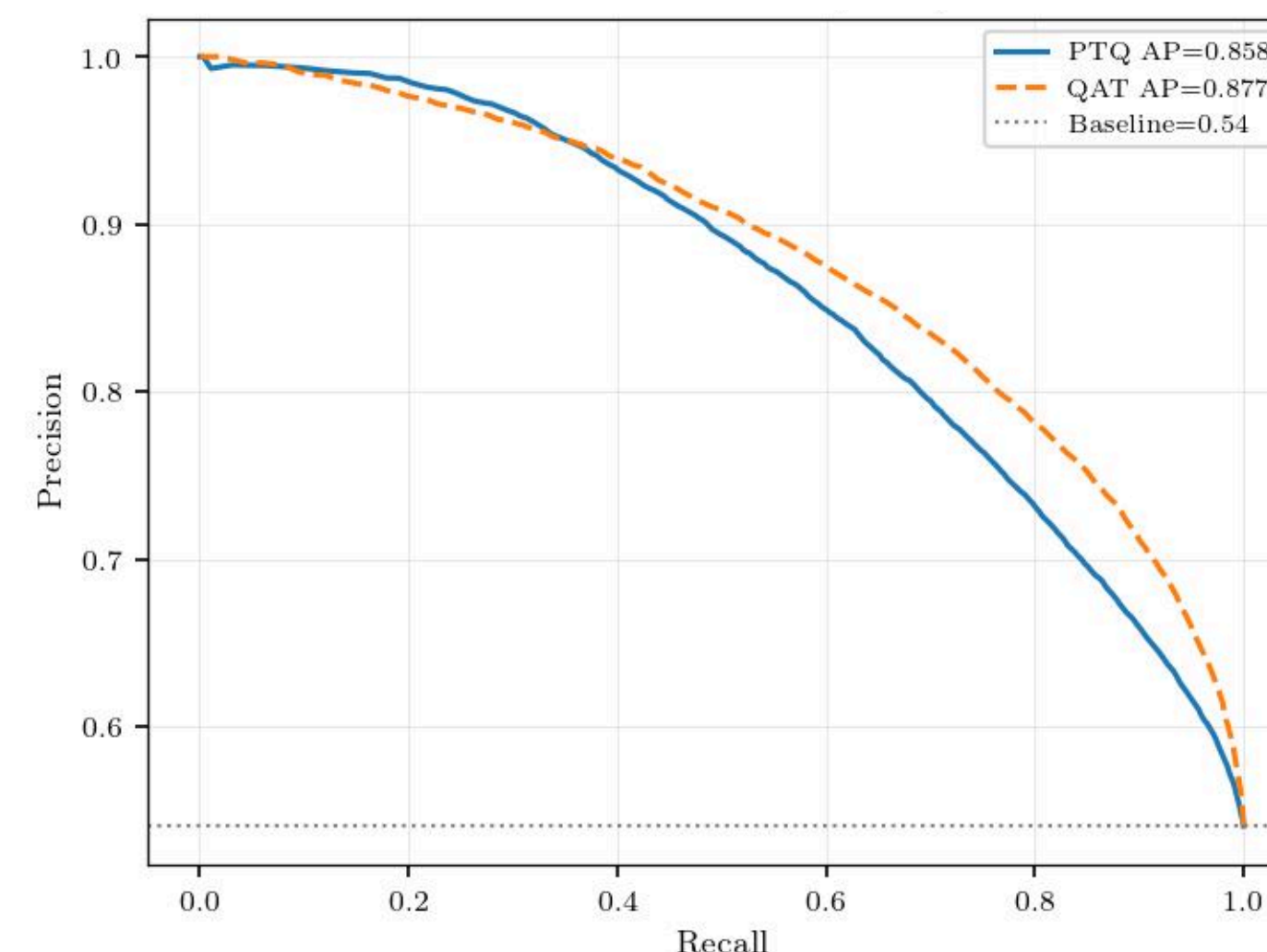
Activation memory on MCUs is the binding constraint, not parameter storage.
The 100 KB tensor arena on external PSRAM accomodates peak intermediate feature map (18,432 elements after the first convolution).

Metrics III: Detection quality

ROC curve



Prec-Rec curve



Evaluation @ 0.6 threshold:

- 48,287 images tested (COCO 2017 dataset)
- Precision: 88.4%
- False alarm rate: 8.9%
- Recall: ~57.5%

Confusion matrix

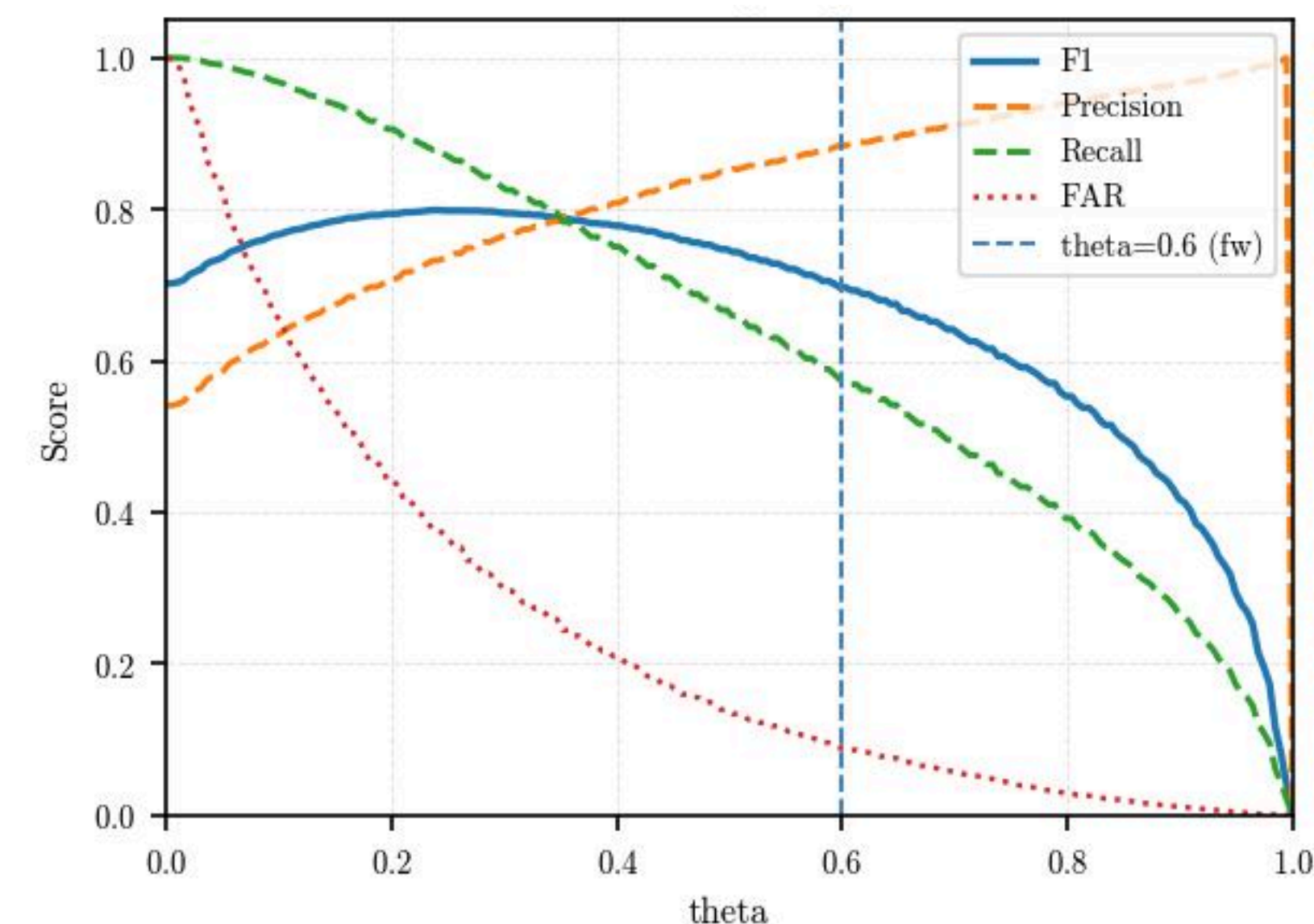
PTQ PTQ th=0.60
F1=0.695 FAR=0.116

True label	Predicted label	
	no_person	person
no_person	19617	2562
person	10851	15257

QAT QAT th=0.60
F1=0.697 FAR=0.089

True label	Predicted label	
	no_person	person
no_person	20210	1969
person	11087	15021

Threshold sweep QAT



Conclusion

Fully on-device inference successfully eliminates cloud dependency, bandwidth drain, and privacy risks while maintaining 88.4% precision

Future work:

- Upgrade to MCUNet and test.
- Re-benchmark on a surveillance-specific dataset.